

SOME PROPERTIES OF BIRTH AND DEATH PROCESSES

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1. Summary

It is shown that for both a discrete time and a continuous time birth and death process (a) the Markov chains imbedded respectively at the moments of birth and moments of death are reverse processes of each other, and (b) the process of times between successive births and the process of times between successive deaths are identically distributed in stationarity. Some applications in queueing theory are discussed.

2. Discrete Case

2.1 Notation

Let $X(0), X(1), X(2), \dots$ be a discrete time birth and death process, i.e. a discrete time Markov chain on $\{0, 1, 2, \dots\}$ with one step stationary transition matrix

$$(1) \quad \begin{vmatrix} r_0 & p_0 & 0 & 0 & \dots \\ q_1 & r_1 & p_1 & 0 & \dots \\ 0 & q_2 & r_2 & p_2 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{vmatrix}$$

We will refer to $X(t)$ as the "population size" at time t .

Some authors refer to this process as a "general random walk" (Parzen (1962), p. 230, Karlin (1966), p. 86). We will suppose that $p_1 > 0, 1 \geq 0$ and $q_1 > 0, 1 \geq 1$ and we write $q_0 = 0$. Easy modifications will make our results applicable in other cases. Define

$$(2) \quad \pi_0 = 1, \pi_n = p_0 p_1 \dots p_{n-1} / q_1 q_2 \dots q_n.$$

It is known (Karlin (1966), p. 88) that the process can be classified as

- (3) transient iff $\sum 1/p_n \pi_n < \infty$
- (4) positive recurrent iff $\sum 1/p_n \pi_n = \infty, \sum \pi_n < \infty$
- (5) null recurrent iff $\sum 1/p_n \pi_n = \infty, \sum \pi_n = \infty$.

Also, in the positive recurrent case the stationary distribution of $\{X(t)\}$ is given by $\{\hat{\pi}_n\}$ where

$$(6) \quad \hat{\pi}_n = \pi_n / \sum \pi_n = \hat{\pi}_0 \pi_n.$$

2.2 The chain imbedded at the moments of births

Let $M_1^B, M_2^B, \dots$ denote the moments of births, i.e.

$$(7) \quad M_1^B = \min \{n > 0 : X(n) > X(n-1)\}$$

$$M_2^B = \min \{n > M_1^B : X(n) > X(n-1)\}, \text{ etc.}$$

Let X_n^B be the population size immediately prior to the n -th birth, i.e.

$$(8) \quad X_n^B = X(M_n^B - 1).$$

Then $X_1^B, X_2^B, \dots$ form a Markov chain; we calculate its transition matrix P . Clearly

$$(9) \quad P_{ij} = 0 \text{ for } j > i + 1.$$

If the population is at size k then the probability of a transition resulting from a death given that a transition occurs is $q'_k = q_k/(q_k + p_k)$ and the probability of a transition resulting from a birth given that a birth occurs is $p'_k = p_k/(q_k + p_k)$. Now the transition $i \rightarrow j \leq i + 1$ of the $\{X_n^B\}$ process occurs if a birth which brought the population size to $i+1$ is followed by $i+1-j$ successive deaths followed by a birth; the probability of this is $q'_{i+1} q'_i \dots q'_{j+1} p'_j$. Hence

$$(10) \quad P_{ij} = p'_{i+1} \quad \text{for } j = i+1$$

$$q'_{j+1} q'_j \dots q'_{i+1} p'_j \quad \text{for } 0 \leq j \leq i+1.$$

A straightforward application of Theorem 4.1 p. 74 of Karlin (1966) shows that (3) remains valid for $\{X_n^B\}$. We show that it is

$$(11) \quad \text{positive recurrent iff } \sum p_n \pi_n < \infty$$

$$(12) \quad \text{null recurrent iff } \sum 1/p_n \pi_n = \infty, \sum p_n \pi_n = \infty.$$

To this end, let us compute

$$(13) \quad \sum_{n=0}^{\infty} p_n \pi_n P_{nj} = \sum_{n=j-1}^{\infty} p_n \pi_n p'_j \prod_{a=j+1}^{n+1} q'_a$$

Denoting by d_n the n -th term of this sum we have the relation

$$(14) \quad d_n = \beta_{n-1} d_{n-1} / (1 + \beta_n)$$

where

$$\beta_n = p_{n+1}/q_{n+1}.$$

$$\text{Hence } (1 + \beta_n) d_n = \beta_{n-1} d_{n-1}.$$

Iterating backwards we get

$$(15) \quad \sum_{n=j-1}^k d_n = (1 + \beta_{j-1}) d_{j-1} - \beta_k d_k \leq (1 + \beta_{j-1}) d_{j-1}.$$

Hence $\sum d_n$ converges and $\beta_k d_k \rightarrow$ some limit, $\gamma \geq 0$, say. Since $(1 + \beta_{j-1}) d_{j-1} = p_j \pi_j$, (15) yields

$$(16) \quad \sum_{n=0}^{\infty} p_n \pi_n p_{nj} = p_j \pi_j - \gamma \leq p_j \pi_j.$$

Now, if the chain is positive recurrent it follows from Theorem 3.2 p. 133 of Karlin (1966) that $\sum p_n \pi_n < \infty$. On the other hand, if $\sum p_n \pi_n < \infty$ then $p_k \pi_k > 0$; thus $\beta_k d_k \leq p_k \pi_k > 0$; hence $\gamma = 0$ and (16) together with Theorem 3.1 p. 132 of Karlin (1966) shows that the chain is positive recurrent. (11) and (12) therefore follows.

Further, in the positive recurrent case the stationary distribution of $\{X_k^B\}$ is $\{\rho_n\}$ where

$$(17) \quad \rho_n = p_n \pi_n / \sum p_n \pi_n.$$

2.3 The chain imbedded at moments of deaths

Let $M_1^D, M_2^D, \dots$ denote the moments of deaths, i.e.

$$(18) \quad M_1^D = \min \{n > 0 : X(n) < X(n-1)\}$$

$$M_2^D = \min \{n > M_1^D : X(n) < X(n-1)\}, \text{ etc.}$$

Let X_n^D denote the population size immediately after the n -th death, i.e.

$$(19) \quad X_n^D = X(M_n^D).$$

Then $X_1^D, X_2^D, \dots$ form a Markov chain; its transition matrix Q can be computed by an argument similar to that given for P ; we find

$$(20) \quad Q_{ij} = \begin{cases} 0 & \text{for } j < i-1 \\ q'_i & \text{for } j = i-1 \\ p'_j p'_{i+1} \dots p'_j q'_j + 1 & \text{for } j \geq i. \end{cases}$$

Now note that

$$(21) \quad Q_{ij} = p_j \pi_j p_{ji} / p_i \pi_i$$

from which follows that $\{X_n^B\}$ and $\{X_n^D\}$ are reverse processes of each other (Karlin (1966) p. 136). The classification of this process is the same as that of the previous and they have identical stationary distributions in the positive recurrent case.

2.4 The processes of times between births and times between deaths

In this paragraph we assume the basic process to be double-ended and stationary, i.e. the process is

$$(22) \quad \dots, X(-1), X(0), X(1), \dots$$

with $X(t)$ distributed according to $\{\hat{\pi}_n\}$ and transitions governed by (1). Define

$$(23) \quad U_n^B = 1 \text{ if } X(n) > X(n-1) \\ 0 \text{ otherwise}$$

$$(24) \quad U_n^D = 1 \text{ if } X(n) < X(n-1) \\ 0 \text{ otherwise.}$$

Then $\{U_n^B\}$ and $\{U_n^D\}$ are stationary processes (Breiman (1968) p. 105); in fact they are identically distributed. To prove this we begin by showing that

$$(25) \quad P(U_0^B = 1) = P(U_0^D = 1).$$

On the left we have

$$(26) \quad P(X(0) > X(-1)) = \sum_{k=0}^{\infty} \hat{\pi}_k p_k = \hat{\pi}_0 \sum_{k=1}^{\infty} \pi_{k-1} p_{k-1}$$

On the right we get

$$P(X(0) < X(-1)) = \sum_{k=0}^{\infty} \hat{\pi}_k q_k = \hat{\pi}_0 \sum_{k=1}^{\infty} \pi_{k-1} p_{k-1}$$

where we use the relation $\pi_k q_k = \pi_{k-1} p_{k-1}$. (25) follows. Hence also

$$(27) \quad P(U_0^B = 0) = P(U_0^D = 0).$$

Next we show that

$$(28) \quad P(U_0^B = 1, U_1^B = 1) = P(U_0^D = 1, U_1^D = 1).$$

On the left we have

$$P(X(-1) < X(0) < X(1)) = \hat{\pi}_0 \sum_{k=0}^{\infty} \pi_k p_k p_{k+1}$$

On the right we get

$$P(X(-1) > X(0) > X(1)) = \hat{\pi}_0 \sum_{k=2}^{\infty} \pi_k q_k q_{k-1}$$

and on noting that $\pi_k q_k q_{k-1} = \pi_{k-2} p_{k-2} p_{k-1}$ (28) follows. From this and (25) and (27) we get

$$P(U_0^B = i_0, U_1^B = i_1) = P(U_0^D = i_0, U_1^D = i_1)$$

for $i_i = 0, 1$. Continuing in this fashion it follows that $\{U_n^B\}$ and $\{U_n^D\}$ are identically distributed.

We have

$$M_1^B = \min \{n > 0 : U_n^B = 1\}$$

$$M_2^B = \min \{n > M_1^B : U_n^B = 1\}, \text{ etc.}$$

with similar expressions for $M_1^D, M_2^D, \dots$ in terms of $\{U_n^D\}$. It follows that $\{M_n^B\}$ and $\{M_n^D\}$ are identically distributed. Further, if $T_1^B = M_1^B, T_2^B = M_2^B - M_1^B, \dots$ denote the times between successive births and if $T_1^D, T_2^D, \dots$ are similarly defined for deaths, then $\{T_n^B\}$ and $\{T_n^D\}$ are identically distributed.

From 6.38 p. 123 of Breiman (1968) we also conclude that $T_1^B, T_2^B, \dots$ form a stationary process conditionally given $U_0^B = 1$ and that $T_1^D, T_2^D, \dots$ form another stationary process conditionally given $U_0^D = 1$; it is clear that these conditional processes are identically distributed under their respective conditions.

From this we also have

$$(29) \quad E(T_1^B | U_0^B = 1) = E(T_1^D | U_0^D = 1) = 1/P(U_0^B = 1)$$

and using (26) and (17) this can be expressed as $\rho_0/p_0\hat{\pi}_0$.

We will now study the common distribution of T_n^B given $U_0^B = 1$; to this end define

$$(30) \quad g(k, m) = P\{X(0) \geq X(1) \geq \dots \geq X(k-1) < X(k) | X(0) = m\} \text{ for } k = 1, 2, \dots$$

By conditioning on the moment at which the first death occurs we get the recursive relation

$$(31) \quad g(k, m) = \sum_{a=1}^{k-1} r_m^{a-1} q_m g(k-a, m-1) + r_m^{k-1} p_m.$$

Also

$$(32) \quad g(1, m) = p_m$$

$$(33) \quad g(k, 0) = r_0^{k-1} p_0.$$

These boundary conditions and (31) determine $g(k, m)$ uniquely for all k and m . We have

$$(34) \quad P(T_1^B = k | U_0^B = 1) = P(T_1^B = k, X(0) > X(-1)) / P(U_0^B = 1).$$

The denominator has been calculated above; the numerator equals

$$(35) \quad \begin{aligned} \sum_{m=0}^{\infty} P(X(-1) = m, X(0) = m+1, T_1^B = k) \\ = \sum_{m=0}^{\infty} \hat{\pi}_m p_m P(X(0) \geq X(1) \geq \dots \geq X(k-1) < X(k) | X(0) = m+1) \end{aligned}$$

$$= \sum_{m=0}^{\infty} \tilde{\pi}_m p_m g(k, m+1)$$

In principle we have found the common distribution of interbirth times. The generating function of this distribution can be obtained somewhat more explicitly as follows. From (34) and (35)

$$\begin{aligned} (36) \quad & \sum_{k=1}^{\infty} s^k P(T_1^B = k | U_0^B = 1) \\ &= \sum_{m=0}^{\infty} \pi_m p_m \bar{g}_{m+1}(s) / \sum \pi_m p_m \end{aligned}$$

where

$$(37) \quad \bar{g}_m(s) = \sum_{k=1}^{\infty} s^k g(k, m).$$

From (31), multiplying by s^k and summing while interchanging the order of summation we get

$$(38) \quad \bar{g}_m(s) = s(p_m + q_m \bar{g}_{m-1}(s)) / (1 - sr_m)$$

From (33)

$$\bar{g}_0(s) = sp_0 / (1 - sr_0).$$

Iterating (38) we get

$$(39) \quad \bar{g}_m(s) = \sum_{k=0}^m sp_k / (1 - sr_k) \prod_{a=k+1}^m sq_a / (1 - sr_a).$$

This together with (36) determines the generating function of the interbirth times distribution.

3. The continuous time case

3.1 Notation

Let $\{Y(t), t \geq 0\}$ be a continuous time birth and death process, i.e. a Markov process on the states $\{0, 1, 2, \dots\}$ with stationary transition probabilities

$$(40) \quad P_{ij}(t) = P[Y(t+s) = j | Y(s) = i]$$

satisfying the conditions

$$\begin{aligned} (41) \quad & P_{i, i+1}(h) = \lambda_i h + o(h), & i \geq 0 \\ & P_{i, i-1}(h) = \mu_i h + o(h), & i \geq 1 \\ & P_{i, i}(h) = 1 - (\lambda_i + \mu_i)h + o(h), & i \geq 0 \\ & P_{ij}(0) = \delta_{ij} \end{aligned}$$

where $\mu_0 = 0$ and we will suppose that $\mu_i > 0$ for $i > 0$ and $\lambda_i > 0$ for $i \geq 0$. $Y(t)$ will again be called the population size at time t . We may take it that all sample functions of $\{Y(t)\}$ are right continuous, that is $Y(t+) = Y(t)$ for all t . Also we assume that $\{\lambda_i\}$, $\{\mu_i\}$ completely determine the transition probabilities of the process (see Karlin and McGregor (1957)).

If we redefine

$$(42) \quad \pi_0 = 1, \pi_n = \lambda_0 \lambda_1 \dots \lambda_{n-1} / \mu_1 \dots \mu_n$$

then the classification of the process is exactly as specified by (3), (4) and (5) with p_n replaced by λ_n ; also the stationary distribution is obtained by normalizing $\{\pi_n\}$ (Karlin and McGregor (1957)).

3.2 Imbedded chains

Let $S_1, S_2, \dots$, denote the successive transition (birth or death) moments; let $S_1^B, S_2^B, \dots$ denote the successive birth moments and $S_1^D, S_2^D, \dots$ the death moments. Then we can write $S_n^B = S_{M_n^B}$ and $S_n^D = S_{M_n^D}$ for M_n^B and M_n^D appropriately defined.

Further, let $\{X(n)\}$ be the population size at the transition moments, i.e. $X(n) = Y(S_n)$. Then $\{X(n)\}$ forms a discrete time birth and death process with parameters

$$q_i = \mu_i / (\mu_i + \lambda_i), p_i = \lambda_i / (\mu_i + \lambda_i), r_i = 0.$$

The population size X_n^B at the moment immediately before the n -th birth is given by

$$X_n^B = Y(S_n^B-) = Y(S_{M_n^B-}) = Y(S_{M_{n-1}^B}) = X(M_n^B - 1).$$

Hence $\{X_n^B\}$ may be thought of as imbedded in $\{X(n)\}$. Similarly the population sizes immediately after deaths $X_n^D = Y(S_n^D)$ may be thought of as imbedded in $\{X(n)\}$ exactly as in the discrete case. It follows that the discussion and conclusions of the discrete time case carry over to the continuous time case.

3.3 Processes of times between births and deaths

In this paragraph we assume that the process $Y(t)$ is double-ended and stationary, i.e. $Y(t)$ is distributed according to $\{\pi_n\}$ for all t , $-\infty < t < \infty$.

We will use a construction of Kingman (1961) to reduce the problems on the identical distributions of $\{S_n^B\}$ and $\{S_n^D\}$ to the equivalent ones in the discrete time case.

Assume that the $\{\lambda_i\}$ and $\{\mu_i\}$ sequences are bounded and let $a > \lambda_i + \mu_i$ for all i . Let $\{X(n)\}$ be a double-ended stationary discrete time birth and death process with parameters

$$(43) \quad q_i = \mu_i/a, p_i = \lambda_i/a, r_i = 1 - (\mu_i + \lambda_i)/a.$$

Note that the π -parameters of this process is the same as that of the Y -process. Let $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ be independent Poisson processes with $EN_1(t) = EN_2(t) = at$ and $N_1(0) = N_2(0) = 0$ and let these processes be independent of the X -

process. Let $N(t) = N_1(t)$ for $t \geq 0$ and $N(t) = -N_2(-t)$ for $t < 0$. Then the Markov process $\{X(N(t)), -\infty < t < \infty\}$ is a stationary continuous time birth and death process satisfying (41) and may therefore be identified with the Y -process, i.e. we may take

$$(44) \quad Y(t) = X(N(t)).$$

Now we show that $\{S_n^B\}$ and $\{S_n^D\}$ are identically distributed. Let M_n^B and M_n^D be defined for the X -process as in (7) and (18). Then

$$P(S_1^B \leq t) = \sum_{k=1}^{\infty} P(S_1^B \leq t, M_1^B = k).$$

On the event $(M_1^B = k)$ S_1^B is the time it takes the Poisson process $N_1(t)$ to count up to k which has a gamma distribution with parameter k and is independent of M_1^B ; denoting the distribution function of this gamma distribution by $G_k(t)$, we have

$$(45) \quad P(S_1^B \leq t) = \sum_{k=1}^{\infty} G_k(t) P(M_1^B = k).$$

A similar argument shows that

$$P(S_1^D \leq t) = \sum_{k=1}^{\infty} G_k(t) P(M_1^D = k).$$

Since M_1^B and M_1^D are identically distributed, we conclude that S_1^B and S_1^D are identically distributed. The argument easily extends to the processes $\{S_n^B\}$ and $\{S_n^D\}$.

Put

$$T_1^B = S_1^B, T_2^B = S_2^B - S_1^B, \dots$$

and

$$T_1^D = S_1^D, T_2^D = S_2^D - S_1^D, \dots$$

Then $\{T_n^B\}$ and $\{T_n^D\}$ are identically distributed. We will now show that conditionally given a birth at time 0 $\{T_n^B\}$ is stationary and that conditionally given a death at 0 $\{T_n^D\}$ is stationary, their conditional stationary distributions being equal. More precisely,

$$(46) \quad \begin{aligned} \lim_{h \rightarrow 0} P(T_1^B \leq t_1, \dots, T_n^B \leq t_n \mid Y(-h) < Y(0)) \\ = \lim_{h \rightarrow 0} P(T_1^D \leq t_1, \dots, T_n^D \leq t_n \mid Y(-h) > Y(0)) \\ = \lim_{h \rightarrow 0} P(T_{j+1}^B \leq t_1, \dots, T_{j+n}^B \leq t_n \mid Y(-h) < Y(0)) \end{aligned}$$

for all integers j .

We have

$$(47) \quad \begin{aligned} P(Y(-h) < Y(0)) &= P(X(N(-h)) < X(0)) \\ &= e^{-ah} ah P(X(-1) < X(0)) + R(h) \end{aligned}$$

where $R(h) = \sum_{n=2}^{\infty} e^{-ah} (ah)^n P(X(-n) < X(0))/n! = o(h)$.

Also

$$\begin{aligned}
 (48) \quad & P(T_1^B \leq t, Y(-h) < Y(0)) \\
 &= \sum_{k=1}^{\infty} P(T_1^B \leq t, M_1^B = k, X(N(-h)) < X(0)) \\
 &= \sum_{k=1}^{\infty} G_k(t) P(M_1^B = k, X(N(-h)) < X(0))
 \end{aligned}$$

using the same argument as in the derivation of (45). Further

$$\begin{aligned}
 & P(M_1^B = k, X(N(-h)) < X(0)) \\
 &= \sum_{n=1}^{\infty} e^{-ah} (ah)^n P(M_1^B = k, X(-n) < X(0))/n! \\
 &= e^{-ah} ah P(M_1^B = k, X(-1) < X(0)) + R_k(h)
 \end{aligned}$$

where

$$|R_k(h)| \leq \sum_{n=2}^{\infty} e^{-ah} (ah)^n / n! = o(h)$$

uniformly in k . Substituting in (48), dividing by (47) and letting $h \rightarrow 0$ we have

$$\begin{aligned}
 (49) \quad & \lim_{h \rightarrow 0} P(T_1^B \leq t | Y(-h) < Y(0)) \\
 &= \sum_{k=1}^{\infty} G_k(t) P(M_1^B = k | U_0^B = 1)
 \end{aligned}$$

with U_0^B as defined by (23). A similar argument shows that

$$\begin{aligned}
 (50) \quad & \lim_{h \rightarrow 0} P(T_1^D \leq t | Y(-h) > Y(0)) \\
 &= \sum_{k=1}^{\infty} G_k(t) P(M_1^D = k | U_0^D = 1)
 \end{aligned}$$

and since M_1^D given $U_0^D = 1$ has the same distribution as M_1^B given $U_0^B = 1$ the first part of (46) follows for $n = 1$. By stationarity of $\{M_n^B\}$ given $U_0^B = 1$ (49) is equal to

$$\begin{aligned}
 & \sum_{k=1}^{\infty} G_k(t) P(M_{j+1}^B = k | U_0^B = 1) \\
 &= \lim_{h \rightarrow 0} P(T_{j+1}^B \leq t | Y(-h) < Y(0))
 \end{aligned}$$

from which the second part of (46) follows for $n = 1$. The argument easily extends to arbitrary n .

Finally we compute the Laplace transform of the times between successive births (or deaths) in stationarity, i.e. of (49). Since

$$\int_0^{\infty} e^{-st} dG_k(t) = (a/(a+s))^k$$

this Laplace transform is

$$(51) \quad \sum_{k=1}^{\infty} (a/(a+s))^k P(M_1^B = k | U_0^B = 1) \\ = \sum_{m=0}^{\infty} \pi_m \lambda_m \bar{g}_{m+1}(a/(a+s)) / \sum \pi_m \lambda_m$$

by comparison with (36). In the present case, using (39) and (43),

$$(52) \quad \bar{g}_m(a/(a+s)) = \sum_{k=0}^m \lambda_k / (s + \mu_k + \lambda_k) \prod_{\alpha=k+1}^m \mu_{\alpha} / (s + \mu_{\alpha} + \lambda_{\alpha})$$

Substituting in (51) we get the required Laplace transform (which, of course, does not depend on a).

This expression cannot be simplified much in general, but if $\mu_k = \mu$, constant, considerable simplification is possible. Using the same device as that by means of which (13) was reduced to (16), the Laplace transform becomes

$$(53) \quad \rho_0 \lambda_0 \mu / (s + \lambda_0) + (1 - \rho_0) \mu / (s + \mu)$$

with ρ_0 as defined by (17) (with p_n replaced by λ_n).

The assumption of boundedness of $\{\mu_i\}$ and $\{\lambda_i\}$ made above was necessary only to make the use of Kingman's construction possible; the results are, however, true without this assumption. So far we have only been able to prove this by laborious computations which need not be reported here.

4. Some Queueing Applications

4.1 M/M/1 system

In this case Burke (1956) has shown that the distribution of the time between successive departures is the same as that of the time between successive arrivals (hence exponential) in stationarity. This is a special case of our results since the queue size of the M/M/1 system is a birth and death process.

4.2 M/M/1 system with balking

For the M/M/1 system in which arriving customers join with probability a_i if i customers are ahead, the queue size again forms a birth and death process with parameters $\lambda_i = \lambda a_i$, $\mu_i = \mu$ where λ is the parameter of the Poisson arrival process and $1/\mu$ the expected service time. The main results of this paper have been proved earlier by the authors (1970) in this special case.

It is worth remarking that the simple expression (53) for the Laplace transform of the time between successive departures can be obtained very simply in this case, as follows: A departing customer leaves the queue non-empty with probability $1 - \rho_0$ or empty with probability ρ_0 . In the first case the time until the next departure is the exponential (μ) service time of the next customer, which has Laplace transform $\mu/(s + \mu)$. In the second case it is the sum of the exponentially distributed (λa_0) time until the first joining of the empty queue and the subsequent exponential (μ) service time; the Laplace transform of this time is $\lambda a_0 \mu / (s + \lambda a_0) + \mu / (s + \mu)$. Hence the interdeparture time distribution has Laplace transform

$$(54) \quad (1 - \rho_0)\mu/(s + \mu) + \rho_0\lambda a_0\mu/(s + \lambda a_0)(s + \mu).$$

Clearly the expected interdeparture and interjoining times are both given by

$$(55) \quad \rho_0(1/\mu + 1/\lambda a_0) + (1 - \rho_0)/\mu \\ = 1/\mu + \rho_0/\lambda a_0$$

In conclusion we remark that for the case $a_0 = 1$, $a_i = p$ for $i = 1, 2, \dots$ we have $\rho_0 = 1 - \lambda p/\mu$ so that the expected interjoining (interdeparture) time in this case becomes

$$1/\lambda + (1 - \rho)/\mu.$$

Takamatsu (1963/1964) claims that the distribution of the interjoining time is exponential with this mean for this special choice of the a_i 's. However, in this case (53) does not represent the Laplace transform of an exponential distribution. That Takamatsu's result is in error can be seen by specializing further to the case $a_0 = 1$, $a_i = 0$ for all $i \geq 1$; then the interjoining time must be the sum of the exponentially distributed interarrival time and the exponentially distributed service time which cannot again be exponentially distributed.

4.3 M/M/s system with balking and reneging

(See Ancker and Gafarian (1963) for the formulation of this queueing model.)

In this case the queue size again forms a birth and death process; thus our main results are applicable, i.e. the distribution of the time between successive departures is the same as that of the time between successive joinings in stationarity.

5. REFERENCES

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